SOHAM ASHISH GAONKAR

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EDUCATION

Indian Institute of Technology Gandhinagar, Gujarat, India — July 2023 – Present
B.Tech. in Artificial Intelligence (CPI: **9.3 / 10.00**, **Rank: 1st** in Dept.)
Incoming M.Tech. Research, Indian Institute of Science (IISc), Bangalore (Accepted via Early Admission)
Relevant Coursework: Machine Learning, Computer Vision, Natural Language Processing, Data Science, Databases, Foundations of Multiagent Systems, Mathematical Foundations for AI, Operating Systems, Data Structures & Algorithms.

PUBLICATIONS

[1] **S. Gaonkar**, H. Shekhar, K. Bader, et al. "Deep Learning-based Ultrasound Image Segmentation for Histotripsy Ablation Zone Delineation." *IEEE International Ultrasonics Symposium (IUS 2025)*, Rotterdam, Netherlands. (Accepted)

[2] **S. Gaonkar**, H. Shekhar, K. Bader, et al. "Improved Segmentation of Histotripsy Bubble Clouds Using Modern CNN Architectures and Combined Dice–Focal Loss." *Frontiers in Acoustics.* (Submitted)

RESEARCH EXPERIENCE

Samsung Research Institute, Noida — May 2026 – Present
Research Intern — Next Level Labs, On-device AI Team
- Optimizing Large Vision-Language Models (VLMs) and LLMs for on-device inference on Samsung mobile chipsets, focusing on memory-efficient attention layouts and quantization-aware compression.
- Conducting latency profiling on mobile NPUs/GPUs; identifying bottlenecks in token generation and attention modules.

Soket AI — Jan 2026 – Feb 2026
Research Intern
- Developed Agri-LLM, agricultural domain-specific language model targeted for Indian farming applications.
- Designed LLM-as-a-judge automated evaluation methodologies and generated supervised fine-tuning (SFT) datasets.

CVIG Lab, IIT Gandhinagar — [Code] | May 2025 – July 2025
Summer Research Intern (Advisor: Prof. Shanmuganathan Raman)
- Developed *PanoTo3D*, a single-panorama 3D scene reconstruction and novel view synthesis pipeline. [Interactive Demo]
- Structured depth-based point clouds; synthesized multi-view supervision via inpainting to stabilize geometry learning; trained 3D Gaussian Splatting models for sparse-view inverse rendering.

MUSE Lab, IIT Gandhinagar — [Code] | Jan 2025 – April 2025
Undergraduate Researcher (Advisor: Prof. Himanshu Shekhar; Collaborator: UChicago Radiology)
- Developed DeepLabV3-based models with custom combined Dice–Focal loss to segment histotripsy ablation bubble clouds in clinical ultrasound imagery, improving mean IoU to **83%** (+7% over prior baseline) and accuracy to **97%**.

SELECTED RESEARCH PROJECTS

LLM Compression via Neural Alignment & Layer Merging — [Code] | Oct 2025
Advisor: Prof. Mayank Singh, IIT Gandhinagar
- Investigated representation alignment limits in LLaMA-3 (8B), analyzing task-dependent representation decay.
- Optimized layer merge weights via gradient descent and Bayesian search; implemented Hungarian permutation matching for optimal neuron alignment, achieving **+0.0864 MMLU score improvement** at **12.5% parameter compression**.

SWaT Anomaly Detection & Root Cause Analysis — [Code] | Jan 2026
Industrial Project, IIT Gandhinagar
- Implemented a 15-stage LSTM Autoencoder pipeline to detect anomalies on SWaT industrial testbed dataset. Integrated Bayesian Networks for causal modeling and reconstruction-based root cause analysis (RCA).

JPEG Compression Pipeline on FPGA — [Code] | April 2025
Advisor: Prof. Joycee Mekie, IIT Gandhinagar
- Designed synthesizable Verilog modules and finite state machines (FSMs) on Basys3 FPGA for discrete cosine transform (DCT) and coefficient pruning, achieving 50% compression with PSNR $>$ 30 dB.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Verilog, JavaScript, Java, SQL
Libraries & Frameworks: PyTorch, Hugging Face Transformers, OpenCV, Scikit-learn, Weights & Biases, SciPy
Systems & Tools: Git, Docker, Basys3 FPGA, Xilinx Vivado, VS Code, Linux, FastAPI, React

HONORS & LEADERSHIP

- **Academic Honors:** Academic Excellence Scholarship (2024–25, 2025–26); Department Rank 1 (B.Tech. AI, IITGN).
- **National Rank:** JEE Advanced AIR 2825 (Top 1.5% of 190k); JEE Main AIR 1987 (Top 0.2% of 1M).
- **Academic Service:** Mentor, MA 103 Calculus & Linear Algebra (300+ students); Core Member, ML Club, IITGN.
- **Competitions:** Top 150 teams out of 20,000+ in Amazon ML Challenge 2025; Runner-up in Robotics HackRush.